

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

ASSESSMENT TOOL 2 – Industry Problem Solving Task

Course Code:	ITA05	Course Title:	Computer Vision
CO Assessed:	CO4 – Articulation (combining methods for evaluation) P4	Assessment:	Assessment Tool 2 – Industry Problem Solving Task
Weightage:	20% of CIA	Total Marks:	100
CO4 Statement:	Compare object and pattern recognition techniques for interpreting visual data with spatial and motion characteristics. (BTL5)		
Student Name:	Yoga Madha A	Reg. No.:	192425058
Section / Batch:	B	Date:	20-08-2026

Code of Conduct

/ Yoga Madha A-192425058 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student:



Criteria	Weight age	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning	Marks
Understanding of Industry Problem	20%	Comprehensive understanding of real-world problem and constraints	Good understanding with minor gaps in requirements	Basic understanding; some requirements unclear	Poor understanding; misinterprets problem context	
Application of Engineering Concepts	25%	Applies advanced and relevant concepts effectively to solve the problem	Applies appropriate concepts with minor errors	Limited application of concepts	Incorrect or no application of concepts	
Solution Design	25%	Innovative, practical, and feasible solution aligned with industry standards	Logical and feasible solution with minor gaps	Basic solution with limited feasibility	Unclear or impractical solution	
Use of Tools	15%	Effective use of modern tools, technologies, and real data	Appropriate use of tools with correct implementation	Limited or inconsistent use of tools	Minimal or incorrect use of tools	
Analysis	10%	Thorough analysis with validated results and performance evaluation	Good analysis with reasonable validation	Basic analysis with limited validation	Poor or no analysis	
Professionalism	5%	Highly professional, well-documented, and clearly presented work	Good documentation and presentation	Basic documentation with some gaps	Poor documentation and presentation	

11. Traffic Surveillance Model Selection

A traffic surveillance system must operate reliably in **real-time road environments**. The selected model should satisfy three important requirements:

1. **Accuracy $\geq 90\%$** – The system must correctly detect vehicles and traffic events with high reliability.
2. **Latency ≤ 80 ms** – The model must process each input quickly enough for real-time traffic monitoring.
3. **Works under varying lighting conditions** – The model should remain reliable during daytime, nighttime, shadows, glare, and changing illumination.

Therefore, a model should **not be selected based on accuracy alone**. It must satisfy **all specified constraints simultaneously**.

Given Model Performance

Method	Accuracy	Latency	Accuracy Requirement	Latency Requirement	Overall
Method A	88%	60 ms	Does not satisfy	Satisfies	Reject
Method B	91%	85 ms	Satisfies	Does not satisfy	Reject
Method C	92%	75 ms	Satisfies	Satisfies	Select

Evaluation of Method A

Method A: 88% accuracy and 60 ms latency

- Accuracy = **88%**
- Required accuracy = **90%**
- Therefore, it is **2 percentage points below** the required accuracy.
- Latency = **60 ms**
- Required maximum latency = **80 ms**
- Therefore, it satisfies the latency requirement.

Although Method A is fast, its accuracy is insufficient for a traffic surveillance application.

Decision: Reject Method A.

A traffic monitoring system with only 88% accuracy may produce more false detections or missed vehicles, reducing reliability.

Evaluation of Method B

Method B: 91% accuracy and 85 ms latency

- Accuracy = **91%**
- Required accuracy = **90%**
- Therefore, it satisfies the accuracy requirement.
- Latency = **85 ms**
- Maximum allowed latency = **80 ms**
- Therefore, it exceeds the latency limit by **5 ms**.

Method B provides acceptable accuracy but is too slow for the specified real-time requirement.

Decision: Reject Method B.

The additional 5 ms may appear small, but in a continuous traffic surveillance system, higher latency can reduce real-time responsiveness.

Evaluation of Method C

Method C: 92% accuracy and 75 ms latency

- Accuracy = **92%**
- Required accuracy = **90%**
- It exceeds the requirement by **2 percentage points**.
- Latency = **75 ms**
- Maximum allowed latency = **80 ms**
- It is **5 ms below** the maximum latency limit.

Therefore, Method C satisfies both quantitative constraints.

Decision: Select Method C.

Method C provides the best balance between **detection accuracy and real-time processing speed**.

Engineering Analysis

The selection can be represented using a constraint-based decision:

Accuracy condition:

$$\text{Accuracy} \geq 90\%$$

Latency condition:

$$\text{Latency} \leq 80\text{ms}$$

For each method:

- **Method A:**
(88 < 90) → **Fail**
- **Method B:**
(91 $\geq$ 90), but (85 > 80) → **Fail**
- **Method C:**
(92 $\geq$ 90) and (75 $\leq$ 80) → **Pass**

Thus:

Method C is the only method satisfying all constraints

Varying Lighting Conditions

The given numerical data does not provide separate accuracy measurements for **daylight, low-light, night, glare, or shadows**. Therefore, lighting robustness cannot be proven directly from the provided table.

For an actual industry deployment, Method C should additionally be tested using:

- Bright daylight
- Night-time conditions
- Low-light scenes
- Shadows
- Headlight glare
- Rainy or cloudy conditions
- Different camera angles

A suitable evaluation would compare the accuracy and latency of each method under these conditions.

For example:

Lighting Condition	Required Evaluation
Daylight	Vehicle detection accuracy
Low light	Detection accuracy and false positives
Night	Detection accuracy and latency
Shadows	Missed-detection rate
Glare	False-detection rate
Rain/cloudy	Overall robustness

Since Method C already satisfies the accuracy and latency constraints, it is the **strongest candidate**, subject to validation under varying lighting conditions.

Practical Solution Design

For a real-world traffic surveillance system, **Method C should be selected as the baseline model.**

A practical system can follow this architecture:

Traffic Camera → Image Pre-processing → Method C Detection Model → Vehicle/Traffic Detection → Real-Time Analysis → Monitoring Dashboard/Alert System

The system should continuously monitor:

- Vehicle detection accuracy
- Processing latency
- False-positive rate
- False-negative rate
- Performance under different lighting conditions
- Frames processed per second

If Method C maintains acceptable performance across these conditions, it can be considered suitable for deployment.

Use of Engineering Tools

Modern traffic surveillance systems can use tools such as:

- **Python** – Model implementation and evaluation
- **OpenCV** – Video and image processing
- **TensorFlow/PyTorch** – Deep-learning model development
- **GPU acceleration** – Reducing inference latency
- **Real-time camera/video streams** – Testing practical performance
- **Performance monitoring tools** – Measuring accuracy, latency and FPS

For validation, the model should be tested on a representative traffic dataset containing different road environments and lighting conditions.

Performance Comparison

Criterion	Method A	Method B	Method C
Accuracy	88%	91%	92%
Latency	60 ms	85 ms	75 ms
Accuracy margin	−2%	+1%	+2%
Latency margin	+20 ms	−5 ms	+5 ms
Recommendation	Reject	Reject	Select

Final Justification

Method C is the best choice because it is the **only method that satisfies both mandatory performance constraints.**

- Method A is fast but fails the accuracy requirement.
- Method B is accurate enough but fails the latency requirement.
- Method C achieves **92% accuracy and 75 ms latency**, satisfying both requirements with a reasonable safety margin.

Therefore:

Selected Method = Method C

Final conclusion: For a real-time traffic surveillance system, **Method C should be selected**, provided additional testing confirms that its performance remains stable under varying lighting conditions. This decision demonstrates a balanced engineering approach because it considers **accuracy, latency, robustness, practical deployment, and system reliability**, rather than selecting a model based on a single performance metric.